

Review retrospective:
*the anharmonic `GibbsObservable` witnesses, and an import
the build called*

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Abstract

A soundness review of `Laplace/OneD/AnharmonicGibbsObservableMonomials.lean`: the file that discharges the `Threepoint.GibbsObservable` side conditions for the anharmonic monomial observables x, x^2, x^3 (the witnesses the anharmonic FDT review consumed), plus the $|x|^k e^{-tL}$ integrability lemma the partition-derivative base uses. Result: **a clean fidelity pass with one dead-import removal** — a redundant `IntegralRemainder` import that a full build confirmed inert.

1 Setting

This is the witness factory below the anharmonic FDT/cumulant files: it proves, by dominated differentiation under the integral sign, that the monomial observables satisfy the abstract `GibbsObservable` predicate, so that the FDT identities become unconditional in those side conditions. It also proves the $|x|^k e^{-tL}$ integrability lemma reused by the partition-derivative base.

2 What was reviewed

`integrable_abs_pow_mul_exp_neg_t_anharmonic`, the general `Threepoint.anharmonic_id_gibbsObservable_pow`, and the $k = 1, 2, 3$ specialisations (`_id`, `_mul_self`, `_mul_mul_self`).

3 Fidelity / soundness findings

Sound; no defects. The integrability lemma is the standard coercivity-to-Gaussian domination: $cx^2 \leq L$ gives $e^{-tL} \leq e^{-tcx^2}$, times $|x|^m$, integrable by the Gaussian-times-polynomial lemma. The general `gibbsObservable_pow` supplies the full dominated-differentiation package — $h = 0$ numerator identity and integrability (reduced to the integrability lemma), AE measurability by continuity, the dominator $t e^{t/(2c)} |x|^{k+1} e^{-(t/2)L}$ (the $k = 0$ pointwise bound times $|x|^k$), its integrability (the lemma at exponent $k + 1$, temperature $t/2$), the pointwise `HasDerivAt`, and the `hasDerivAt_integral_of_dominated_loc_of_deriv_le` finish. The three specialisations are function-extensional transports along $x^1 = x$, $x^2 = x x$, $x^3 = x x x$. The reviewer confirmed all four.

4 Simplifications applied

One dead import. `import Laplace.OneD.IntegralRemainder` is referenced by no symbol — `integrable_abs_pow_mul_exp_neg_t.anharmonic` is *defined* here, not imported from it. The reviewer flagged this as “build-arbitrated only”, worrying that the Mathlib lemma `integrable_abs_pow_mul_exp_neg_mul_sq` used in the integrability proof might come *through* `IntegralRemainder`. A full lake build (8306 jobs) stayed green with the import removed, settling it: that lemma is in scope via the kept imports, and no consumer relies on the transitive dependency. `open MeasureTheory` and the other two imports are load-bearing. Signatures byte-identical. GPT affirmed.

5 Disagreements and open questions

None. The reviewer’s optional `simp` compressions were declined as proof-golf.

6 Lessons

- The reviewer’s “maybe this lemma comes from that import” is **exactly the transitive-load-bearing question — and the full build answers it**. GPT correctly refused to call `IntegralRemainder` dead from the text alone, since `integrable_abs_pow_mul_exp_neg_mul_sq` *could* have lived there. The green full build is the adjudication: it does not. This is the same discipline as the `AddSeparable` case (where the build went the *other* way and kept a symbol-less import) — “no symbol referenced” is a hypothesis the build tests, in both directions.

7 Follow-ups

- The remaining anharmonic arc (`AnharmonicGibbsRegularity`, `AnharmonicKappa3Affine`) and the harmonic regularity/observable/perturbed files.